Year 11 · AQA Level 2 Further Maths · Lesson 1

Quadratics: completing the square and the formula

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Today I will learn to...

Goals

1. Complete the square to solve and find turning points

2. Solve quadratics with the quadratic formula

Do now: retrieval practice

4 mins

1 Factorise $x^2 + 7x + 12$

$(x + 3)(x + 4)$

2 Expand $(x - 5)^2$

$x^2 - 10x + 25$

3 Simplify $\sqrt{48}$

$4\sqrt{3}$

4 Solve $(x - 2)(x + 7) = 0$

$x = 2$ or $x = -7$

Objective 1

Complete the square to solve and find turning points

Key idea

Learn

!

$x^2 + bx + c = (x + b/2)^2 - (b/2)^2 + c$. Halve the x-coefficient, square it, subtract it. For $ax^2 + bx + c$, take a out of the x-terms first. $y = (x + p)^2 + q$ has turning point $(-p, q)$.

Objective 1: Complete the square to solve and find turning points

I do: example 1

I do

Write $x^2 + 6x + 2$ in the form $(x + p)^2 + q$

1. Halve 6: $p = 3$
2. $(x + 3)^2 = x^2 + 6x + 9$
3. Subtract 9, then add 2
4. $(x + 3)^2 - 7$

I do: example 2

I do

Write $2x^2 - 12x + 5$ in the form $a(x + b)^2 + c$

1. Take out 2: $2(x^2 - 6x) + 5$
2. $x^2 - 6x = (x - 3)^2 - 9$
3. $2[(x - 3)^2 - 9] + 5$
4. $= 2(x - 3)^2 - 13$

I do: example 3

I do

Solve $x^2 - 4x - 1 = 0$ by completing the square. Give exact answers.

1. $(x - 2)^2 - 4 - 1 = 0$
2. $(x - 2)^2 = 5$
3. $x - 2 = \pm\sqrt{5}$
4. $x = 2 \pm \sqrt{5}$

Mini whiteboards

We do

1. Write $x^2 + 10x + 3$ in the form $(x + p)^2 + q$

Answer: $(x + 5)^2 - 22$

2. Write down the turning point of $y = (x - 4)^2 + 7$

Answer: $(4, 7)$

3. Write $x^2 - 6x + 1$ in the form $(x + p)^2 + q$

Answer: $(x - 3)^2 - 8$

4. Write down the minimum value of $(x + 2)^2 - 5$

Answer: -5

You do: choose your challenge

You do

Bronze

Write $x^2 + 4x + 9$ in the form $(x + p)^2 + q$

Silver

Write $x^2 - 7x + 2$ in the form $(x + p)^2 + q$

Gold

Write $2x^2 + 8x + 1$ in the form $a(x + b)^2 + c$.
State the turning point of $y = 2x^2 + 8x + 1$.

Platinum

Solve $x^2 + 6x + 4 = 0$ by completing the square.
Give exact answers.

You do: answers

Check

Bronze

$(x + 2)^2 + 5$

Silver

$(x - 3.5)^2 - 10.25$

Gold

$2(x + 2)^2 - 7$, turning point $(-2, -7)$

Platinum

$(x + 3)^2 = 5$, so $x = -3 \pm \sqrt{5}$

If you finish early

Extra

Bronze

Write $x^2 + 8x + 20$ in the form $(x + p)^2 + q$

$(x + 4)^2 + 4$

Silver

Write $x^2 - 3x + 1$ in the form $(x + p)^2 + q$

$(x - 1.5)^2 - 1.25$

Gold

Write $3x^2 - 18x + 5$ in the form $a(x + b)^2 + c$

$3(x - 3)^2 - 22$

Platinum

Find the turning point of $y = 2x^2 + 12x + 7$. Is it a maximum or a minimum?

$2(x + 3)^2 - 11$: minimum at $(-3, -11)$

Exam question 1

3 marks

Write $x^2 - 8x + 20$ in the form $(x - a)^2 + b$. Hence explain why $x^2 - 8x + 20 > 0$ for all values of x.

Space for working

Mark scheme: $(x - 4)^2 + 4$ (M1 A1). A square is never negative, so the expression is at least 4 (B1)

Exam question 2

3 marks

Write $3x^2 + 12x - 1$ in the form $a(x + b)^2 + c$

Space for working

Mark scheme: $3(x^2 + 4x) - 1$ (M1). $3[(x + 2)^2 - 4] - 1$ (M1). $3(x + 2)^2 - 13$ (A1)

Objective 1 achieved

Complete the square to solve and find turning points



Objective 2

Solve quadratics with the quadratic formula

✓ Achieved: Complete the square to solve and find turning points

Key idea

Learn

!

For $ax^2 + bx + c = 0$: $x = (-b \pm \sqrt{(b^2 - 4ac)}) \div 2a$. Write down a, b and c with their signs first. Work out $b^2 - 4ac$ on its own, and simplify any surd.

Objective 2: Solve quadratics with the quadratic formula

I do: example 1

I do

Solve $x^2 + 5x + 3 = 0$. Give answers to 2 dp.

1. $a = 1, b = 5, c = 3$
2. $b^2 - 4ac = 25 - 12 = 13$
3. $x = (-5 \pm \sqrt{13}) \div 2$
4. $x = -0.70$ or $x = -4.30$

I do: example 2

I do

Solve $2x^2 - 3x - 4 = 0$. Give answers to 2 dp.

1. $a = 2, b = -3, c = -4$
2. $b^2 - 4ac = 9 + 32 = 41$
3. $x = (3 \pm \sqrt{41}) \div 4$
4. $x = 2.35$ or $x = -0.85$

I do: example 3

I do

Solve $x^2 - 6x + 4 = 0$. Give exact answers in simplest surd form.

1. $b^2 - 4ac = 36 - 16 = 20$
2. $x = (6 \pm \sqrt{20}) \div 2$
3. $\sqrt{20} = 2\sqrt{5}$
4. $x = 3 \pm \sqrt{5}$

Mini whiteboards

We do

1. Solve $x^2 + 3x - 2 = 0$ to 2 dp

Answer: $x = 0.56$ or $x = -3.56$

2. Solve $x^2 + 2x - 2 = 0$ exactly

Answer: $x = -1 \pm \sqrt{3}$

3. Solve $x^2 - 4x - 3 = 0$ exactly

Answer: $x = 2 \pm \sqrt{7}$

4. Solve $2x^2 + 5x + 1 = 0$ to 2 dp

Answer: $x = -0.22$ or $x = -2.28$

You do: choose your challenge

You do

Bronze

Solve $x^2 + 4x + 1 = 0$ to 2 dp

Silver

Solve $2x^2 + x - 5 = 0$ to 2 dp

Gold

Solve $x^2 - 10x + 7 = 0$ exactly

Platinum

A right-angled triangle has sides x, x + 2 and hypotenuse x + 3. Find x to 2 dp.

You do: answers

Check

Bronze

$x = -0.27$ or $x = -3.73$

Silver

$x = 1.35$ or $x = -1.85$

Gold

$x = (10 \pm \sqrt{72}) \div 2 = 5 \pm 3\sqrt{2}$

Platinum

$x^2 + (x + 2)^2 = (x + 3)^2$ gives $x^2 - 2x - 5 = 0$, so $x = 1 + \sqrt{6} = 3.45$

If you finish early

Extra

Bronze

Solve $x^2 + 6x + 2 = 0$ to 2 dp

$x = -0.35$ or $x = -5.65$

Silver

Solve $3x^2 - 2x - 4 = 0$ to 2 dp

$x = 1.54$ or $x = -0.87$

Gold

Solve $x^2 - 8x + 13 = 0$ exactly

$x = 4 \pm \sqrt{3}$

Platinum

A rectangle is $(2x + 1)$ cm by $(x - 1)$ cm and its area is 30 cm^2. Find x to 2 dp.

$2x^2 - x - 31 = 0$, $x = (1 + \sqrt{249}) \div 4 = 4.20$

Exam question 1

3 marks

Solve $3x^2 + 7x - 2 = 0$. Give your answers to 2 decimal places.

Space for working

Mark scheme: a = 3, b = 7, c = −2 (M1). (−7 ± √73) ÷ 6 (M1). x = 0.26 or x = −2.59 (A1)

Exam question 2

3 marks

A rectangle is (x + 3) cm long and x cm wide. Its area is 20 cm². Find x to 2 dp.

Space for working

Mark scheme: $x^2 + 3x - 20 = 0$ (M1). $x = (-3 + \sqrt{89}) \div 2$ (M1). $x = 3.22$ (A1)

Objective 2 achieved

Solve quadratics with the quadratic formula



Real life

Apply

A ball's height after t seconds is $h = 20t - 5t^2 + 2$ metres. When does it hit the ground? Give your answer to 2 dp.

$5t^2 - 20t - 2 = 0$, so $t = (20 + \sqrt{440}) \div 10 = 4.10$ seconds

Extension: stretch yourself

Stretch

1. For which values of k does $x^2 + kx + 9 = 0$ have two different real solutions?

$b^2 - 4ac > 0$: $k^2 - 36 > 0$, so $k < -6$ or $k > 6$

2. Show that the minimum value of $x^2 - 6x + 14$ is 5, and find where it happens.

$(x - 3)^2 + 5$: minimum 5 when $x = 3$

Exit ticket: exam questions

Exit

1. Write $x^2 + 12x + 7$ in the form $(x + p)^2 + q$ **(2 marks)**

2. Solve $x^2 - 3x - 5 = 0$. Give your answers to 2 decimal places. **(2 marks)**

3. Solve $x^2 + 8x + 11 = 0$. Give exact answers in the form $a \pm \sqrt{b}$ **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 $(x + 6)^2 - 36 + 7$ (M1) $= (x + 6)^2 - 29$ (A1) [2]

Q2 $(3 \pm \sqrt{29}) \div 2$ (M1). $x = 4.19$ or $x = -1.19$ (A1) [2]

Q3 $(x + 4)^2 = 5$ (M1). $x = -4 \pm \sqrt{5}$ (A1) [2]

Total: ____ / 6

Today I achieved

- ✔ **Complete the square to solve and find turning points**
- ✔ **Solve quadratics with the quadratic formula**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Write $x^2 - 10x + 30$ in the form $(x + p)^2 + q$ [2]

2. Write $3x^2 - 6x + 7$ in the form $a(x + b)^2 + c$ [3]

3. Solve $x^2 + 7x + 4 = 0$. Give your answers to 2 dp. [3]

4. Solve $x^2 - 4x - 8 = 0$. Give exact answers in the form $a \pm b\sqrt{3}$ [3]

5. The curve $y = x^2 + 2x - 6$ has a turning point. Find its coordinates by completing the square. [2]

Extra practice: Mr Barton Maths: AQA Further Maths → search "Completing the square"